Research Statement

Yujie Liu

1 Research Motivation

I have always been interested in uncovering the fundamental structure of real-world operational systems and using that understanding to improve decision making and system performance. Such systems include service systems, supply chains, computing and communication systems, and healthcare operations, where decisions must often be made with limited resources, uncertain demand, and evolving system states. My research is rooted in applied probability, particularly stochastic modeling and queueing theory, and integrates perspectives from modern sequential learning to examine how operational dynamics and available information shape effective decisions.

In studying these systems, I am especially drawn to simple, canonical models that capture the essential features of an operational problem. One recurring question in my research is: When do simple decision rules perform well, when do they break down, and why? I investigate this question by characterizing fundamental performance limits and examining how decision rules behave under different operating conditions. My goal is to explain the mechanisms behind their success or failure and use these insights to guide policy design. This approach connects rigorous mathematical analysis with operational insights into problems such as scheduling, resource allocation, and congestion management.

2 Research Overview

My current work focuses primarily on queueing systems and sequential learning. In queueing systems, I study how service constraints, finite-time performance requirements [1], and information [4] shape system behavior and effective control policies. In sequential learning, I study how adaptive algorithms allocate samples over time, with a focus on the resulting sampling dynamics and refined characterizations of algorithmic performance [2, 3]. An earlier line of my work studies stochastic model reduction, where I develop lower-dimensional representations that preserve essential probabilistic structure while reducing computational complexity [5, 6, 7]. I describe representative projects from these lines of work below.

2.1. Finite-time minimax bounds and an optimal Lyapunov policy in queueing control. Stability is fundamental in queueing systems, but it does not by itself explain how much congestion a policy creates over a finite operating horizon. This distinction is crucial in nonstationary and time-critical regimes that commonly arise in modern service and computing systems, where transient performance directly affects responsiveness and resource efficiency. In my work [1], we develop a minimax framework for finite-time queueing control that characterizes how expected queue length depends on the operating horizon, scheduling capacity, and arrival and service variability. This framework complements prior work on time-uniform transient upper bounds [8, 9] and steady-state upper and lower bounds [10], the latter of which admit a minimax interpretation. By explicitly accounting for the operating horizon, our framework provides a principled benchmark for evaluating scheduling policies beyond classical long-run stability.

Within this framework, we derive fundamental lower bounds on finite-time queue length and develop *LyapOpt*, a Lyapunov-based scheduling policy that attains optimal finite-time performance under suitable heavy-traffic conditions while preserving classical stability guarantees. We further show that MaxWeight, despite its strong long-run stability properties, can exhibit suboptimal finite-time scaling and incur substantially larger transient backlogs. These results reveal a fundamental gap between stability and finite-time performance, clarify the limitations of classical drift-based scheduling, and provide a principled basis for designing queueing-control policies with rigorous transient guarantees.

2.2. Queue regulation based on waiting-time information disclosure. The delay information available to customers shapes their joining decisions and, in turn, server utilization and social welfare. Queue-length information is easy to provide but serves only as a proxy for expected waiting time. When service times are variable, the same queue length can correspond to substantially different workloads, making queue length less informative about the delay a customer will actually experience. This raises a natural question: is it possible for more precise delay information to improve both server utilization and expected customer utility, and if so, under what conditions?

In my ongoing work [4], we compare queue-length disclosure with exact waiting-time disclosure in strategic $M/M/1$ and $M/G/1$ queues. Our analysis extends the comparison in [11] and addresses the question raised there of how service-time variability affects system behavior under different disclosure regimes. Under waiting-time disclosure, customers follow a workload-threshold rule that aligns joining decisions with realized delay. In the $M/M/1$ model, we show that system load shapes the relative performance of the two disclosure regimes. Under suitable conditions, waiting-time disclosure can achieve a win–win outcome for the server and customers that queue-length disclosure cannot. Extending the analysis to $M/G/1$ queues, we find that service-time variability can change both the structure of customer decisions and the relative performance of the two regimes. Deterministic queue-length thresholds may no longer characterize equilibrium joining behavior, whereas the workload-threshold structure under waiting-time disclosure is preserved. Moreover, some performance orderings established in the $M/M/1$ model can reverse. These results illustrate how delay information shapes strategic behavior and how disclosure can serve as an operational tool for regulating congestion and improving welfare.

2.3. Sampling Allocation of LinUCB. Traditional analyses of linear bandits focus primarily on regret, which measures cumulative performance loss but does not directly characterize how samples are allocated across actions over time. Understanding this allocation reveals how information accumulates, provides a basis for refined regret analysis and statistical inference, and may inform the design of more efficient learning algorithms. In my work [2] and ongoing work [3], we study the sampling allocation of LinUCB and its implications for regret and statistical inference. This research extends the study of UCB sampling behavior in classical multi-armed bandits [12, 13] to the linear setting, where the geometry of the action set and the shared parameter structure shape how information is acquired across actions.

In [2], we analyze LinUCB in the small-gap regime, which captures the difficult instances underlying worst-case regret lower bounds. We show that, in this regime, its empirical sampling allocation converges to the set of D-optimal designs, revealing that LinUCB, although designed for regret minimization, is also asymptotically efficient for parameter estimation in the D-optimality sense. This characterization enables refined regret analysis and asymptotically valid statistical inference under adaptive sampling. In my ongoing work [3], we study the fluid-limit dynamics of LinUCB across different gap regimes, uncovering phase transitions in its sampling behavior. Together, these results provide a structural understanding of adaptive sampling that supports refined performance analysis and may inform future algorithm design.

2.4. Minimal representation of phase-type distributions. An earlier line of my research concerns stochastic model reduction through the minimal representation of phase-type distributions. Phase-type distributions are widely used in stochastic modeling because they combine substantial modeling flexibility with tractable Markovian structure, making them particularly useful in queueing and inventory systems [14]. Their computational complexity, however, grows directly with the number of phases in the representation. This motivates the minimal representation problem: for a given phase-type distribution, what is the smallest number of phases needed to represent it exactly? Despite its fundamental role in reducing model complexity, this remains a longstanding problem in applied probability.

My work develops structural and computational approaches to this problem. In [7], we use polyhedral cone theory to characterize minimal representations for an important class of phase-type distributions. In [6], we establish a connection between minimal representations and a constrained inverse eigenvalue problem,

leading to a nonconvex optimization formulation and an alternating-minimization algorithm. Building on these results, in [5], we derive necessary and sufficient conditions for general phase-type distributions with repeated and complex poles to admit a minimal representation. Together, these results connect stochastic model reduction with matrix analysis and optimization, and provide a systematic approach to reducing representation complexity while preserving the underlying distribution exactly.

3 Future Research Plans

Going forward, I plan to build on my work in queueing systems and sequential learning to address decision-making challenges in stochastic operational systems. I am particularly interested in developing simple and implementable decision policies with strong performance guarantees, and in understanding how information and learning should be incorporated into their design. I outline two directions that I plan to pursue.

Design and control of stochastic service systems. Building on my work on finite-time queueing control [1], I plan to study how system design and operational control can be coordinated to improve the performance of stochastic service systems. My current work shows that MaxWeight can exhibit suboptimal finite-time scaling for certain highly asymmetric scheduling sets, highlighting the importance of accounting for resource structure in policy design. One interesting question is how to jointly optimize resource configurations, which determine the scheduling set, and scheduling policies to improve performance over a finite operating horizon under a fixed resource budget. Here, resource configurations are chosen before operations begin, while scheduling decisions adapt to the evolving system state. My goal is to understand how resource configurations and scheduling policies interact and to develop simple, implementable design and control rules with rigorous finite-horizon performance guarantees.

Sequential learning for operational decision making. Drawing on my work on sequential learning and queueing systems, I plan to study how learning can be integrated into operational decisions when demand, service, or reward parameters are unknown. Unlike in standard stochastic bandit models, exploration can create congestion or deplete resources, with consequences that persist through the system dynamics. Estimation errors can also propagate through operational decisions and affect subsequent system performance. I aim to understand how these effects should guide the allocation of exploration resources and the balance between the future value of information and its operational costs. Of particular interest is how learning policies should adapt to the evolving system state and remaining operating horizon. My goal is to develop simple, implementable policies with rigorous guarantees that connect sampling allocation and estimation accuracy to operational performance.

References

[1] **Yujie Liu**, Vincent Y. F. Tan, and Yunbei Xu. Finite-time minimax bounds and an optimal Lyapunov policy in queueing control. Forthcoming in *Operations Research*.

[2] **Yujie Liu**, Vincent Y. F. Tan, and Yunbei Xu. Sampling allocation of LinUCB: Optimal design limits in the small-gap regime. To appear in *ACM SIGMETRICS 2027*.

[3] **Yujie Liu**, Vincent Y. F. Tan, and Yunbei Xu. How LinUCB allocates samples: A fluid-limit analysis under local gaps. Work in progress.

[4] **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. Queue regulation based on waiting-time information disclosure. Manuscript in preparation.

[5] **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. A non-convex optimization approach of searching algebraic degree phase-type representations for general phase-type distributions. *Advances in Applied Probability*, 57(3):1101–1137, 2025.

[6] **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. On substochastic inverse eigenvalue problems with the corresponding eigenvector constraints. *SIAM Journal on Matrix Analysis and Applications*, 45(3):1689–1719, 2024.

[7] **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. Minimal positive realizations for linear systems with two-different-pole transfer functions. *European Journal of Control*, 69:100720, 2023.

[8] Devavrat Shah, John N. Tsitsiklis, and Yuan Zhong. On queue-size scaling for input-queued switches. *Stochastic Systems*, 6(1):1–25, 2016.

[9] Jiaming Xu and Yuan Zhong. Improved queue-size scaling for input-queued switches via graph factorization. *Advances in Applied Probability*, 52(3):798–824, 2020.

[10] Devavrat Shah, Neil Walton, and Yuan Zhong. Optimal queue-size scaling in switched networks. *The Annals of Applied Probability*, 24(6):2207–2245, 2014.

[11] Pengfei Guo and Paul Zipkin. Analysis and comparison of queues with different levels of delay information. *Management Science*, 53(6):962–970, 2007.

[12] Anand Kalvit and Assaf Zeevi. A closer look at the worst-case behavior of multi-armed bandit algorithms. In *Advances in Neural Information Processing Systems*, volume 34, pages 8807–8819, 2021.

[13] Koulik Khamaru and Cun-Hui Zhang. Inference with the upper confidence bound algorithm. arXiv preprint arXiv:2408.04595, 2024.

[14] Marcel F. Neuts. *Matrix-Geometric Solutions in Stochastic Models: An Algorithmic Approach*. The Johns Hopkins University Press, 1981.